Project Initialization and Planning Phase

Date	1 July 2026
Team ID	SWTID-2026-1614
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform socio-economic policy simulation using machine learning, boosting analysis speed and accuracy. It tackles workflow inefficiencies, offering policy planners, researchers, and students a streamlined, interactive environment to run “what-if” scenarios. Key features include a machine learning-based regression model and real-time tier classification.

Project Overview	
Objective	To build an intelligent Human Development Index (HDI) prediction system that leverages machine learning to analyze key socio-economic parameters (Life Expectancy at birth, Mean Years of Schooling, Gross National Income per capita) and predict the optimal HDI score, maximizing planning efficiency.
Scope	The project covers data collection and preprocessing, training a regression model on historical socio-economic data, and deploying a web-based tool that provides real-time HDI projections, target simulation modeling, and baseline profiling for researchers and policy planners.
Problem Statement	
Description	Policy planners and researchers often lack access to a reliable, interactive, and data-driven way to estimate how target indicators (health, education, income) translate to overall developmental scores, forcing them to rely on static spreadsheets or custom programming environments.
Impact	Solving this will help analysts make informed, data-backed policy recommendations, model developmental objectives, and gain better insight into indicator dependencies for national development planning.
Proposed Solution	
Approach	Employing a supervised machine learning regression model (Linear Regression) trained on historical socio-economic indicators (Life Expectancy, Schooling, GNI) to predict the HDI score, served through a modern, responsive web-based form interface.
Key Features	- ML-based HDI prediction engine trained on historical socio-economic data. - Interactive country search dropdown that auto-populates baseline variables for 195 countries. - Glassmorphism input dashboard with manual parameter override for custom target simulation. - Real-time classification of predicted HDI into UNDP-aligned developmental tiers (Low, Medium, High, Very High).

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (dual/quad-core); no GPU required – model is a lightweight regression estimator.
Memory	RAM specifications	4–8 GB
Storage	Disk space for data, models, and logs	< 1 GB (CSV dataset + serialized model files)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE	Jupyter Notebook / VS Code; deployed on Render/Vercel
Data		
Data	Source, size, format	Kaggle/UNDP Human Development Index dataset, csv format containing historical indicator records for 195 nations.